TIER II ADOPTION NOTICE

IEEE-ISTO MOSA-IF-S-001, "Common Payload Interface Standard – Command and Data Handling," was released on July 14, 2020 (as SMC-SIS-001) and was adopted on October 15, 2023 for use by the Department of Defense (DoD) and other space systems. This standard was revised in 2024 for use with Ethernet (MOSA-IF-S-002) rather than Serial RapidIO for high-speed data transfer across multiple payloads. NOTE: The use of MOSA-IF-S-002 is preferred by the industry contractors who participated in its development.

Proposed changes for MOSA-IF-S-001 may be submitted to the IEEE-ISTO Space Systems Modular Open Systems Approach (MOSA) Interface Standards Alliance via an electronic Standard Change Request form [https://interoperabilityalliance.org/resources/scr-form]. Information regarding this technical standard, including copies of the latest revision, may be obtained via the electronic request form on the MOSA Alliance Website [https://interoperabilityalliance.org/store]. This standard is approved for limited distribution, and requestors must be employees of a U.S. Government Agency Contractor.

- IEEE-ISTO MOSA-IF-S-001 specifies an architecture for achieving a standardized bus to payload command and data handling (C&DH) interface and requires the use of SpaceWire and Serial RapidIO. It represents a first incremental delivery of a complete standardized bus to payload interface for on-orbit integration. A major program of record has implemented MOSA-IF-S-001 into their system design for hosted payload interfaces.
- The Standard specifies C&DH interface functions and capabilities including management of high-speed data transfer across any number of payloads. The standard assumes the use of a conforming Centralized Payload Manager (CPM) unit. However, the standardized functions can be implemented within any contractor's existing payload interface units.
- An accompanying Guide (MOSA-IF-G-001) provides recommendations on how to implement a conforming Centralized Payload Manager Unit.
- The standard can be applied to ground-based bus-to-payload C&DH integration or on-orbit bus-to-payload C&DH integration.
- This standard is available in .pdf file format or as a Model-Based Systems Engineering (MBSE) model in Cameo Systems Modeler.

Custodians:
Air Force – 19

Adopting Activity:
Air Force – 19
(MOSS-2024-006)

Reviewing Activities:
OSD – DTIC, SE, SO
Navy – EC, NM
Air Force – 01, 05, 10, 13, 19, 20, 24, 33, 70, 71, 84, 85, 110, 114, 170, 171, 184
DLA – DA, DH
Other – DC5, DI, MDA, NRO, NA, NIST

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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